M&AE 5070 - Flight Dynamics

Chapter 7 - Lateral equilibrium: side slip and turns

Spring 2026

In this chapter, we cover two specific cases of lateral equilibrium: steady turns and side slip straight flight, along with the necessary control surfaces positions.

1 Lateral equations at equilibrium

Equilibrium is reached when the derivative of the state vector is zero. For lateral motion, the state vector contains the roll velocity p, the yaw velocity r, the side-slip angle β, and the roll/bank angle ϕ:

- **R** - Equation for the roll moment, around $\vec{x_b}$:

$$C_\ell = C_{\ell_\beta}\beta + C_{\ell_p}\frac{p\ell}{V} + C_{\ell_r}\frac{r\ell}{V} + C_{\ell_{\delta_\ell}}\delta_\ell + C_{n_{\delta_n}}\delta_n = 0 \tag{1}$$

- **Y** - Equation for the yaw moment, around $\vec{z_b}$:

$$C_{n_=}C_{n_\beta}\beta + C_{n_p}\frac{p\ell}{V} + C_{n_r}\frac{r\ell}{V} + C_{n_{\delta_\ell}}\delta_\ell + C_{n_{\delta_n}}\delta_n = 0 \tag{2}$$

- **LF** - Balance of forces along $\vec{y_b}$: weight projection onto $\vec{y_b}$; aero force along $\vec{y_b}$ (lateral)

$$\underbrace{mV\left(r\cos\alpha_e - p\sin\alpha_e\right)}_{\text{inertial forces}} = mg\sin\phi\cos\theta_e + \frac{1}{2}\rho SV^2\left(C_{Y_\beta}\beta + C_{Y_{\delta_n}}\delta_n\right) \tag{3}$$

- **Kϕ** - Kinematic equation for the roll attitude ϕ around $\vec{x_b}$:

$$p = -\tan\theta_e(r\cos\phi + q_e\sin\phi) \tag{4}$$

Since there are two lateral controls, δ_ℓ and δ_n, two additional equations/constraints are needed to fully determine the equilibrium state. Those additional constraints will be defined in the context of specific flight configurations: turns and balanced side-slip flight.

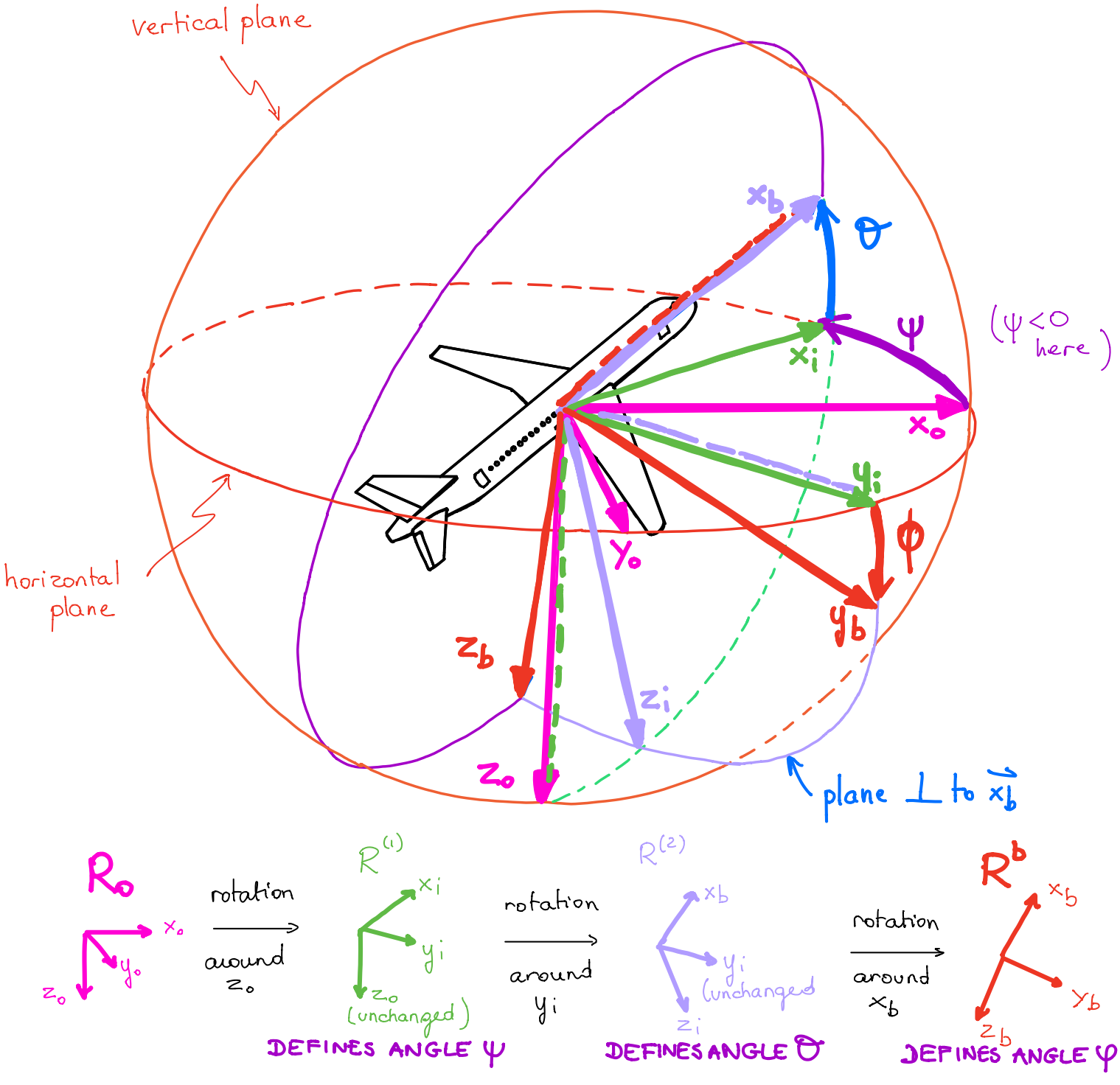
6 variables: p, r, β, ϕ, δ_ℓ, δ_n <=> 4 balance equations + 2 additional constraints to fully close our system.

2 Axes, angular velocities and kinematic equations

Earth Frame of reference : R_o

Aircraft Frame of reference : R_b

↳ Switch from R_o to R_b through 3 successive rotations:

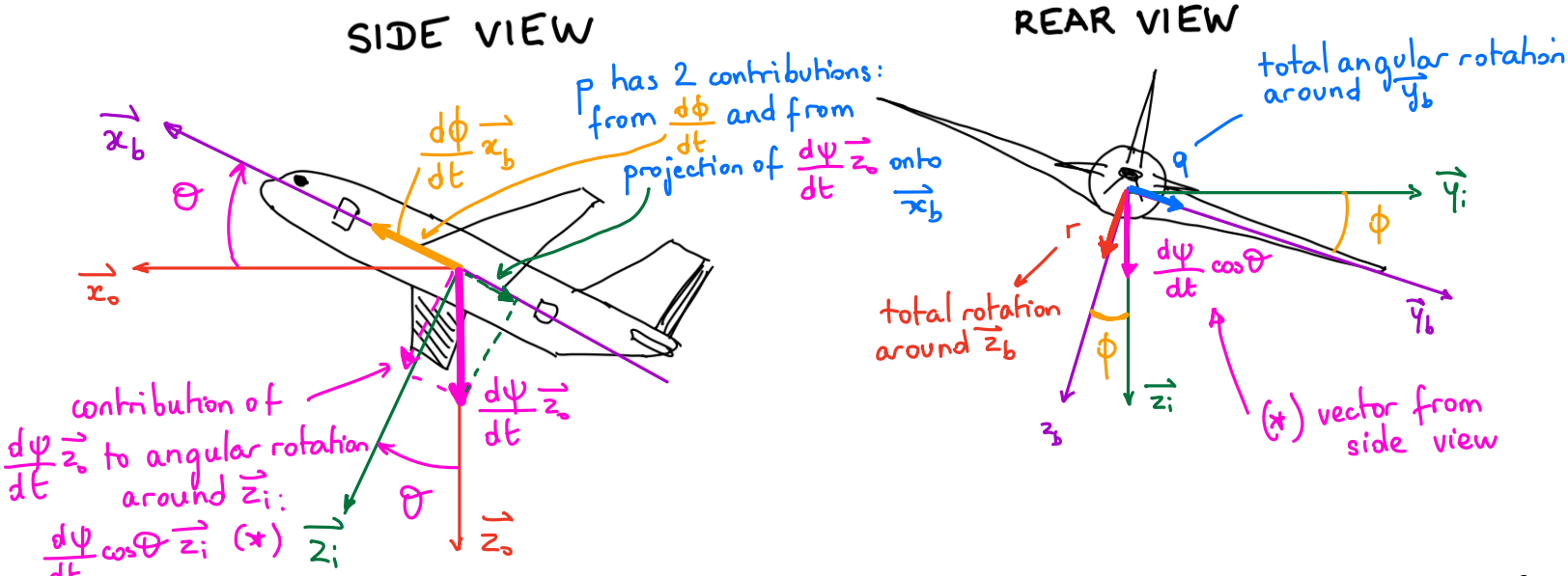


The relevant variable when looking at turns is the aircraft's angular velocity vector $\vec{\Omega}$ **with respect to the Earth**. This vector can be expressed as function of its components in the reference frame of the aircraft, or as function of the derivative of the state angles ψ, θ, and ϕ:

$$\vec{\Omega} = \vec{\Omega}^b = \begin{pmatrix} p \\ q \\ r \end{pmatrix} = p\vec{x}_b + q\vec{y}_b + r\vec{z}_b \qquad (R^b)$$

$$= \frac{d\phi}{dt}\vec{x}_b + \frac{d\theta}{dt}\vec{y}_i + \frac{d\psi}{dt}\vec{z}_o \qquad (\vec{x}_b, \vec{y}_i, \vec{z}_o)$$

Question: how are (p, q, r) related to $\frac{d\phi}{dt}$, $\frac{d\theta}{dt}$, $\frac{d\psi}{dt}$?



From side view:

p (total angular velocity around $\vec{x}_b$) $= \frac{d\phi}{dt} - \frac{d\psi}{dt}\sin\theta \Rightarrow$

$$\boxed{\frac{d\phi}{dt} = p + \frac{d\psi}{dt}\sin\theta}$$

From rear view: contributions to angular rotation around $\vec{z}_i$:

(*) term — projection of $r\vec{z}_b$ onto $\vec{z}_i$

$$\boxed{\frac{d\psi}{dt}\cos\theta = r\cos\phi + q\sin\phi}$$

projection of $q\vec{y}_b$ onto $\vec{z}_i$

accounting for all contributions to angular rotation around $\vec{y}_i$:

$$\boxed{\frac{d\theta}{dt} = q\cos\phi - r\sin\phi}$$

2 Steady turns at constant altitude

- Aircraft trajectory belongs to horizontal plane
- turns ≡ aircraft changes heading (angle ψ)
  ↳ aircraft has a VERTICAL angular velocity $\vec{\Omega} = \begin{pmatrix} 0 \\ 0 \\ \Omega_t \end{pmatrix}$ of magnitude Ω_t

no contribution on horizontal plane

along $\vec{z}_0$

We can express its components in the aircraft's frame of reference R_b using the expressions we just derived relating $\frac{d\psi}{dt}$, $\frac{d\phi}{dt}$, and $\frac{d\theta}{dt}$, and p, q, and r:

If $\vec{\Omega} = \Omega_t \vec{z}_0$, what are the corresponding values of p, q & r?

Starting from expressions in previous page, and using the fact that

$$\begin{cases} \frac{d\phi}{dt} = \frac{d\theta}{dt} = 0 & (\text{no contribution of } \vec{\Omega} \text{ on horizontal plane}) \\ \frac{d\psi}{dt} = \Omega_t & (\vec{\Omega} \text{ aligned with } \vec{z}_0) \end{cases}$$

$$\begin{cases} p = -\Omega_t \sin\theta & (1) \\ \Omega_t \cos\theta = r\cos\phi + q\sin\phi & (2) \\ 0 = q\cos\theta - r\sin\phi & (3) \end{cases} \Rightarrow q = r\frac{\sin\phi}{\cos\phi}$$

substituting q into (2):

$$\Omega_t \cos\theta = r\cos\phi + r\frac{\sin^2\phi}{\cos\phi}$$

$$\Omega_t \cos\theta \cos\phi = r\underbrace{(\cos^2\phi + \sin^2\phi)}_{=1}$$

(*)
$$\begin{cases} p = -\Omega_t \sin\theta \\ q = \Omega_t \cos\theta \sin\phi \\ r = \Omega_t \cos\theta \cos\phi \end{cases}$$

technically, an horizontal turn is associated with non-zero pitch, roll AND yaw angular velocities!

2.1 Simplifying assumptions

- typical rotation rates Ω_t : 0.05 to 0.1 rad/s if $\phi \approx 30°$
  - ↳ we can assume that Ω_t^2 is small
- Assume that α, θ, β are small
- ϕ is NOT small, but $\phi \approx 20°$
  - ↳ replacing $\sin\phi$ by ϕ introduces a 2% error
  - ↳ " $\cos\phi$ by 1 " 6% error
  - ↳ $\Longrightarrow$ we will do it!
- decouple δ_n and δ_ℓ AND ignore δ_n, δ_ℓ coupling with $\vec{Y}$

$$C_{\ell_{\delta_n}} = C_{n_{\delta_\ell}} = 0 \quad \text{AND} \quad C_{Y_{\delta_\ell}} = C_{Y_{\delta_n}} \approx 0$$

- (+) is approximated as

$$\begin{cases} p \approx 0 \\ q \approx 0 \\ r \approx \Omega_t \end{cases}$$

The simplified equations for steady turn at constant altitude become:

- R $\quad C_\ell = 0 = C_{\ell_\beta}\beta + C_{\ell_r}\Omega_t \frac{\ell}{V} + C_{\ell_{\delta_\ell}}\delta_\ell$ ($\Omega_t = r$)
- Y $\quad C_n = 0 = C_{n_\beta}\beta + C_{n_r}\Omega_t \frac{\ell}{V} + C_{n_{\delta_n}}\delta_n$
- LF $\quad mV\Omega_t = mg\phi + \frac{1}{2}\rho S V^2 C_{Y_\beta}\beta$
- $k\phi$ $\quad p = 0$

+ 2 equations/constraints $\begin{cases} r = \Omega_t \\ \text{TBD} \end{cases}$

The last equation determines what kind of turn is performed.

2.2 Turn with zero side-slip

$\beta = 0$

This is what pilots consider a "coordinated" turn: achieves good performances, low drag by cancelling out side-slip. However, this turn requires the use of both the stick and the pedals, and is therefore a bit more complicated to do.

Reminder : we are looking for the controls to MAINTAIN a steady turn, not get INTO the turn.

- Supplemental equation: $\beta = 0$
- **R**

$$\delta_\ell = -\frac{C_{\ell_r}}{C_{\ell_{\delta_\ell}}} \Omega_t \frac{\ell}{V} \quad \Rightarrow \quad \delta_\ell > 0$$

($C_{\ell_r} > 0$, $C_{\ell_{\delta_\ell}} < 0$)

- **Y**

$$\delta_n = -\frac{C_{n_r}}{C_{n_{\delta_n}}} \Omega_t \frac{\ell}{V} \quad \Rightarrow \quad \delta_n < 0$$

($C_{n_r} < 0$, $C_{n_{\delta_n}} < 0$)

- **LF**

$$\phi = \frac{V}{g}\Omega_t = \frac{V}{g}\frac{V}{R_t} = \frac{mV^2}{R_t}\frac{1}{mg}$$

R_t: curvature radius; $\frac{mV^2}{R_t}$: centrifugal force; mg: weight

Comments:

To maintain a right turn ($\Omega_t > 0$)

- $\delta_\ell > 0$: STICK TO THE LEFT (opposite to turn)
- $\delta_n < 0$: PEDAL TO THE RIGHT ("into" the turn)

- Stick and pedal cancel ROLL and YAW moments by compensating the influence of the yaw velocity $r = \Omega_t$
- the banking compensates the centrifugal force with the component of the weight on the wing

2.3 Steady turn without rudder - "operational" turn

This is a more practical turn done on all faster aircraft. It only involves the stick, which simplifies the procedure.

- Supplemental equation: $\delta_n = 0$ no pedal, no rudder
- **R**

$$\delta_\ell = \Omega_t \frac{\ell}{V} \frac{C_{\ell_\beta} C_{n_r} - C_{\ell_r} C_{n_\beta}}{C_{n_\beta} C_{\ell_{\delta_\ell}}}$$

($C_{\ell_\beta}<0$, $C_{n_r}<0$, $C_{\ell_r}>0$, $C_{n_\beta}>0$)

- **Y**

$$\beta = -\Omega_t \frac{\ell}{V} \frac{C_{n_r}}{C_{n_\beta}}$$

($C_{n_r}<0$, $C_{n_\beta}>0$) ⟸ non-zero side-slip

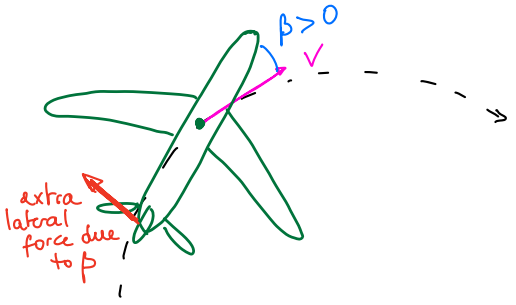
- **LF**

$$\phi = \frac{\Omega_t}{mg}\left(mV + \frac{1}{2}\rho S V^2 C_{Y_\beta} \frac{C_{n_r}}{C_{n_\beta}} \frac{\ell}{V}\right)$$

($C_{Y_\beta}<0$, $C_{n_r}<0$, $C_{n_\beta}>0$)

Comments:

- Sign of $\beta \equiv$ sign of Ω_t
  - ↳ for a right turn, nose of the aircraft goes from left to right AND wind comes from inside of the turn



- Side slip compensates influence of yaw velocity on yaw moment (Y equation)
  - ↳ we can say that δ_n controls side-slip : $\delta_n \Longleftrightarrow \beta$ (just like δ_m was controlling α in longitudinal motion)

- For banking, we again need $\phi \neq 0$ to compensate for:
  - * centrifugal force
  - * AND extra lateral force coming from $\beta \neq 0$

$$\phi_{\text{stick only}} > \phi_{\text{pedal + stick}}$$

- The position of the stick depends on the sign of the numerator:
  - ↳ expression is the same as the one controlling stability of spiral mode

Stick "inside the turn" for **STABLE SPIRAL MODE**

Stick "outside the turn" for **UNSTABLE SPIRAL MODE**

Application: The pilot of a transport aircraft tells a glider pilot that they typically handle turns with the stick alone when on a holding pattern, before landing. The glider pilot is surprised by that, because they tend to use both pedals and stick. Help the transport aircraft pilot justifying their choice. The holding pattern velocity is around $V = 60$ m/s, the turn rate, 360°in one minute, the mean aerodynamic chord, $\ell = 6$m, and the distance between the center of gravity and the vertical stabilizer, $d_{VS} = 20$m.

Steady turn, no rudder : $\delta_n = 0$ $\longrightarrow$ side slip $\beta \neq 0$!

Objective: show that β is small $\Rightarrow$ small effect of side slip

From equation: $$\beta = -\Omega_t \frac{\ell}{V} \frac{C_{n_r}}{C_{n_\beta}}$$

lateral force coeff due to side slip

From previous chapter : $$C_{n_\beta} = -\frac{d_{VS}}{\ell} C_{Y_\beta}$$

$$C_{n_r} = +\left(\frac{d_{VS}}{\ell}\right)^2 C_{Y_\beta}$$

CORRECTION FROM CLASS : ⊕ sign

$$\beta = -\Omega_t \frac{\ell}{V}\left(\frac{d_{VS}}{\ell}\right)^2 \frac{1}{\left(\frac{d_{VS}}{\ell}\right)} = \Omega_t \frac{d_{VS}}{V} \approx 2^\circ$$

small indeed!

°/sec

m/s

Need to convert

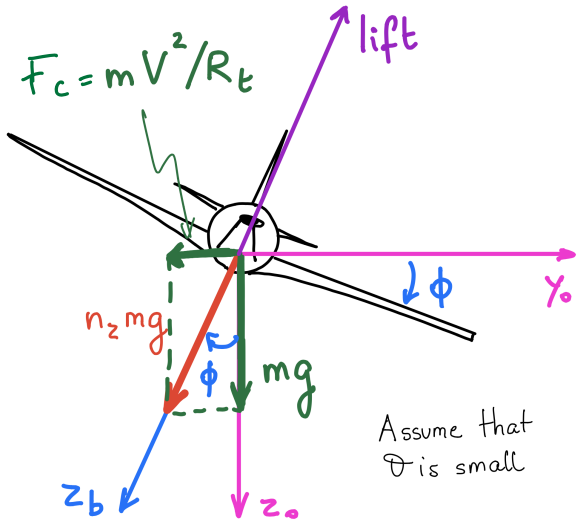
3 Load factor

3.1 Simplified considerations

In the previous equations, we saw that during a classical turn, the aircraft banks on one side ($\phi \neq 0$): in this case, a longitudinal load factor appears, which perturbs the $\boxed{\mathbf{S}}$ equation. In this section, we provide a definition to the load factor in the context of steady turns. It will help re-define some of the characteristics of a turn:

- Rotation rate Ω_t: allows to evaluate how long an aircraft will take to accomplish a given change in heading, $\Delta\psi$.
- Curvature radius or turn radius R_t: allows to evaluate the "volume" an aircraft needs to turn.

During a steady turn with zero side-slip ("coordinated" turn, schematic below), there are two mass-related forces: the weight mg, and the centrifugal force, mV^2/R_t.



The vectorial sum of those two is **by definition of the load factor, equal to the product of the load factor n_z by the weight.** The projection of the vector $mg\overrightarrow{n_z}$ on the vertical direction is equal to the weight:

$$n_z mg \cos\phi = mg \tag{5}$$

or,

$$n_z = \frac{1}{\cos\phi} \tag{6}$$

The load factor during a turn is only function of the bank angle, independently of the aircraft and flight conditions (but assumes that the turn is a "coordinated" turn). A more thorough definition of the load factor is provided next.

3.2 General definition

The governing equations of motion are written

$$m\overrightarrow{\Gamma}_{\text{abs},G} = m\frac{d\overrightarrow{V}^o}{dt} = \overrightarrow{F_{\text{aero}}} + \overrightarrow{F_{\text{prop}}} + m\overrightarrow{g} \tag{7}$$

The mass contribution can be isolated:

$$0 = m\left(\overrightarrow{g} - \frac{d\overrightarrow{V}^o}{dt}\right) + \overrightarrow{F_{\text{aero}}} + \overrightarrow{F_{\text{prop}}} \tag{8}$$

$$0 = \overrightarrow{F_{\text{mass}}} + \overrightarrow{F_{\text{aero}}} + \overrightarrow{F_{\text{prop}}} \tag{9}$$

This mass force is equal to the sum of the weight and the inertia force, which we can re-write introducing the load factor vector $\overrightarrow{n}$:

$$\overrightarrow{F_{\text{mass}}} = \overrightarrow{n}mg = m\left(\overrightarrow{g} - \frac{d\overrightarrow{V}^o}{dt}\right) \tag{10}$$

from which

$$\overrightarrow{n} = \frac{\overrightarrow{g} - \overrightarrow{\Gamma}_{\text{abs},G}}{g} \tag{11}$$

This is the definition of the load factor. We can also write it as:

$$\overrightarrow{n} = -\frac{\overrightarrow{F_{\text{aero}}} + \overrightarrow{F_{\text{prop}}}}{mg} \tag{12}$$

Coming back to the turn with no side-slip problem (figure above). The propulsion force does not appear (along fuselage axis), so only the aerodynamic lift appears, which is equal and opposite to the load factor vector multiplied by the weight. Because it is carried by $\overrightarrow{z}_b$, this vector is often referred to by its component along this axis, n_z, only, which is (confusingly enough) also call load factor.

3.3 Practical consideration

The load factor is an elegant way to translate the forces to which an aircraft and its pilot/passengers are subjected. For example, in a turn with a 60°bank angle, $n_z = 2$, which means that the wing structure must be able to handle twice its weight.

During the preliminary design phase, it is convenient to refer to a maximum load factor. For example, for a civil aircraft, the typical max load factor is $n_{z,0} = 2.5$. It means that the pilot will be able to operate the aircraft at this load factor without risk of damage. Of course, the design is such that rupture occurs at a larger load factor which we call "extreme": $n_{z,ex}$, obtained from the max load factor through a multiplicative safety factor:

$$n_{z,ex} = 1.5n_{z,0} = 3.75 \tag{13}$$

A civil transport aircraft therefore breaks around $4g$. This provides a much sufficient safety margin, even in unusual meteorological events. If you are experiencing heavy turbulence at $1.5g$, you are already flying around if you are not buckled up, yet the aircraft is completely within its normal use conditions, well below the safety margin.

For a fighter jet, the extreme load factor is larger, around $10g$ depending on the type of mission. It means that for a fighter jet around 10 tons, the aircraft structure must be designed to handle a lift force of 100 tons. Such high load factors are linked to the capability of the pilot to resist them. Without going too much into the details, we can give a few orders of magnitude to clarify things. When a fighter jet pilot slowly increase his/her load factor, for example between 0.5 g/s and 1 g/s, they will experience physiological limits that start by a tunnel vision effect. If they maintain the load around $4g$ for more than 5s, they will experience a gray-out. Around $5g$, this moves toward a blackout, where the pilot cannot see anymore, but remains conscious. After that, they will lose consciousness. Again, those are only order of magnitude, a lot depends on the pilot itself. Anti-g suits can help tremendously, since they counter the main cause of gray-out/blackout, the conflict between blood pressure and force linked to the acceleration: if the heart cannot pump blood to the brain efficiently, the brain does not work properly. With them, a pilot may be able to handle very high gs for short amount of time, for example, an ejection at $10g$, but which lasts less than a second.



Figure 1: F-35 during a tight turn: the pilot experiences a sever load factor *[www.f35.com]*

3.4 Useful formulas

We already established the relationship between n_z and ϕ:

$$n_z = \frac{1}{\cos\phi} \tag{14}$$

Let's look at the other key quantities, by projecting $mg\overrightarrow{n_z}$ on the horizontal:

$$n_z mg \sin\phi = m\frac{V^2}{R_t} = F_c \tag{15}$$

$$n_z g \sin\phi = \Omega_t V \tag{16}$$

The centrifugal force is $F_c = m\frac{V^2}{R_t}$, and the second equation has been obtained by using the relationship between R_t, Ω_t and V:

$$R_t \Omega_t = V \tag{17}$$

Using Eq. 14, we have

$$\sin\phi = \frac{\sqrt{n_z^2 - 1}}{n_z} \tag{18}$$

The centrifugal force becomes

$$F_c = mg\sqrt{n_z^2 - 1} \tag{19}$$

and the rotation rate is expressed as:

$$\Omega_t = \frac{g}{V}\sqrt{n_z^2 - 1} = \frac{g}{V}\tan\phi = \frac{F_c}{mV} \tag{20}$$

and the inverse of curvature/turn radius R_t is:

$$\frac{1}{R_t} = \frac{\Omega_t}{V} = \frac{g}{V^2}\sqrt{n_z^2 - 1} = \frac{g}{V^2}\tan\phi \tag{21}$$

Application: A fighter jet pilot does a steady tight turn (with zero side-slip) with a high load factor. He wants to increase the rotation rate Ω_t, but keeping the load factor constant. Ω_t is the derivative of the bearing angle ψ, or, adjusting the units, the number of rotations per minute. To do this, he needs:

(A) decrease his velocity while keeping ϕ constant, then the curvature radius of his turn will decrease.

(B) decrease his velocity while increasing ϕ, then the curvature radius of his turn will decrease.

(C) increase his velocity while increasing ϕ, then the curvature radius of his turn will increase.

(D) increase his velocity while decreasing ϕ, then the curvature radius of his turn will increase.

(E) decrease his velocity while decreasing ϕ, then the curvature radius of his turn will decrease.

$$n_z = \frac{1}{\cos\phi} \longrightarrow \phi \text{ must stay constant}$$

$$R_t \Omega_t = V \longrightarrow R_t = \frac{V_t}{\Omega_t} \longrightarrow R_t \text{ decreases.}$$

$$\Omega_t = \frac{g}{V}\sqrt{n_z^2 - 1} \longrightarrow \Omega_t \text{ increases} \Rightarrow V \text{ must decrease}$$

Application: Turns at constant altitude. Consider a turn made without side-slip. Demonstrate that for this turn to occur at constant altitude and constant velocity, the pilot necessarily need to adjust throttle. What aerodynamic or engineering limitations can you foresee for steady turns that will impact turn characteristics (turn radius R_t, rotation rate Ω_t, velocity during turn etc.)?

[S] Before the turn : $mg = \frac{1}{2}\rho S V^2 C_z$ $(n_z = 1)$

During the turn : $n_{z_t} mg = \frac{1}{2}\rho S V_t^2 C_{z_t}$

↑ load factor > 1

Assuming constant velocity : $V = V_t \Rightarrow n_{z_t} = \frac{C_{z_t}}{C_z} > 1$

$\boxed{C_{z_t} > C_z}$

$\Rightarrow \delta_m$ has to be adjusted too!

[P] $F - D = mg \cancel{\sin\gamma}^{0}$ before turn

$F_t - D_t = 0$ during the turn

$D_t = \frac{1}{2}\rho S V_t^2 C_{x_t}$

$C_{x_t} = C_{x_0} + k C_{z_t}^2$

$C_{z_t} > C_z \Rightarrow D_t > D$

* if δ_x not increased, altitude ↓
* if δ_m is not touched : $V_t > V$ velocity will be higher during turn.

Need to increase throttle!

Limitations on turn performance (Ω_{max}, R_{min})

↳
- n_z below safe value
- $F < F_{max}$
- $C_{z_t} < C_{z_{max}}$